Logistic Regression

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Maximum Likelihood Estimation

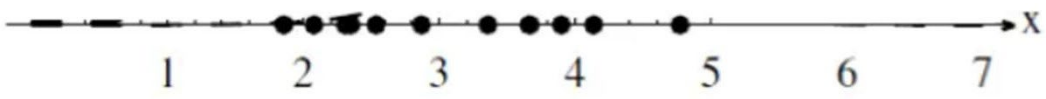
Goal: estimate parameters of a distribution from a dataset $\mathcal{D} = \{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(N)}\}$

- $\mathcal{D}$ contains N independent, identically distributed (i.i.d.) training samples.

We need to determine $\boldsymbol{\theta}$ given $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(N)}\}$

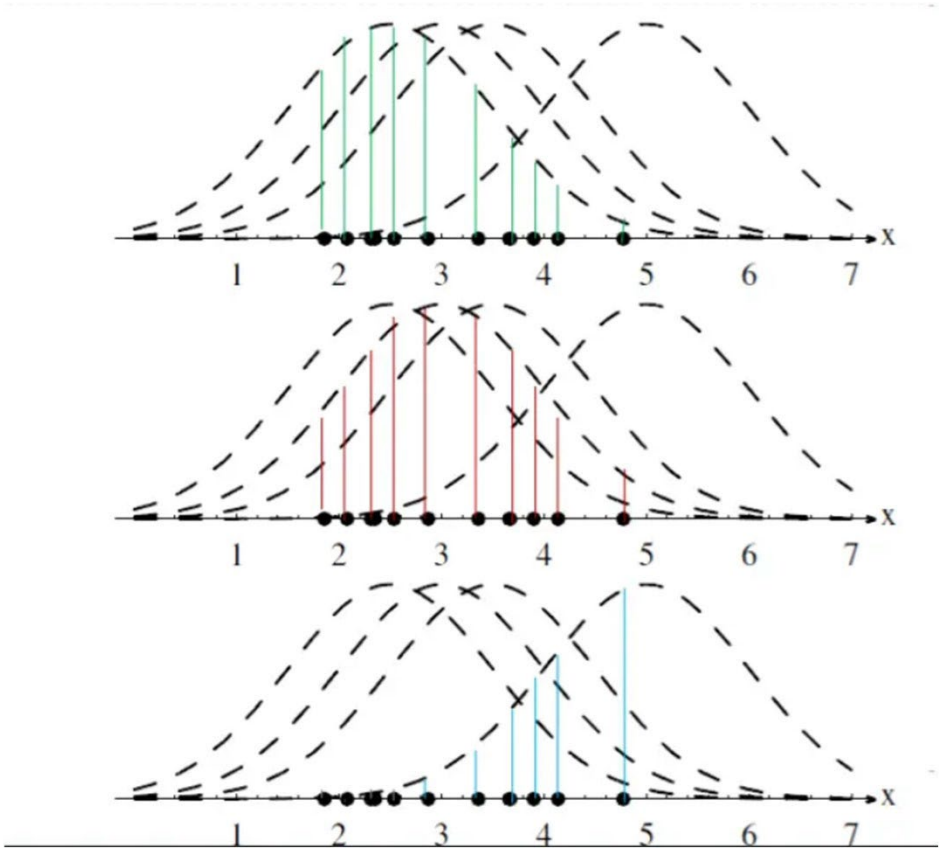
- How to represent $\boldsymbol{\theta}$?

Example



$$P(x|\mu) = N(x|\mu, 1)$$

Example



ML estimation

Maximum-likelihood estimation (MLE) is a method of estimating the parameters of a statistical model given data.

Likelihood is the conditional probability of observations $\mathcal{D} = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(N)}\}$ given the value of parameters $\boldsymbol{\theta}$

- Assuming i.i.d. observations:

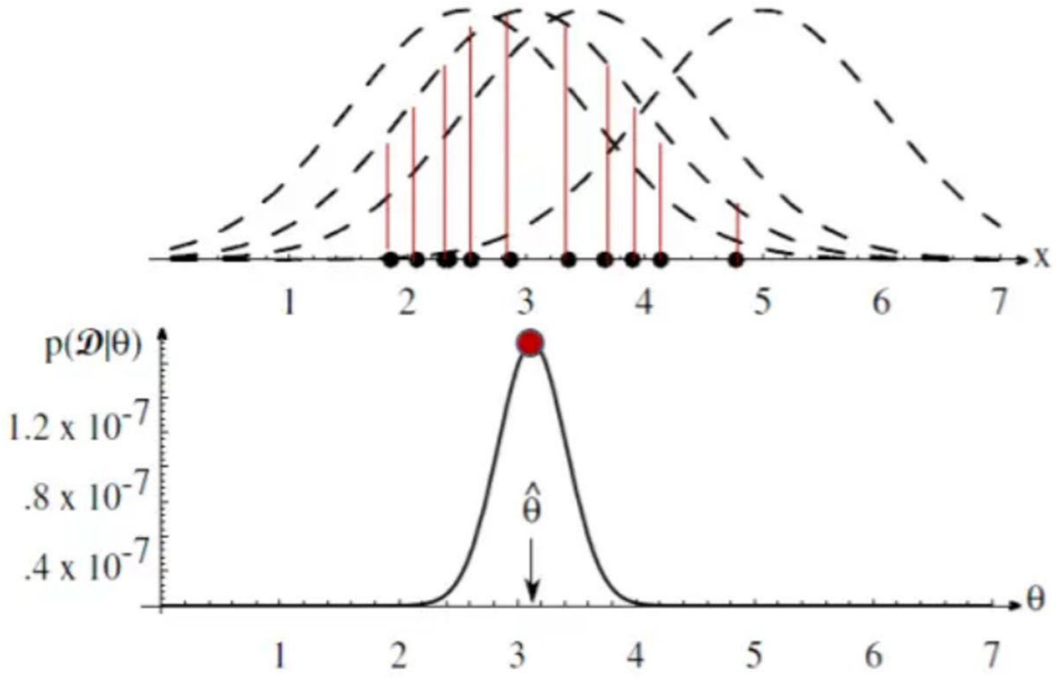
$$p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^{N} p(\boldsymbol{x}^{(i)}|\boldsymbol{\theta})$$

likelihood of $\boldsymbol{\theta}$ w.r.t. the samples

Maximum Likelihood estimation

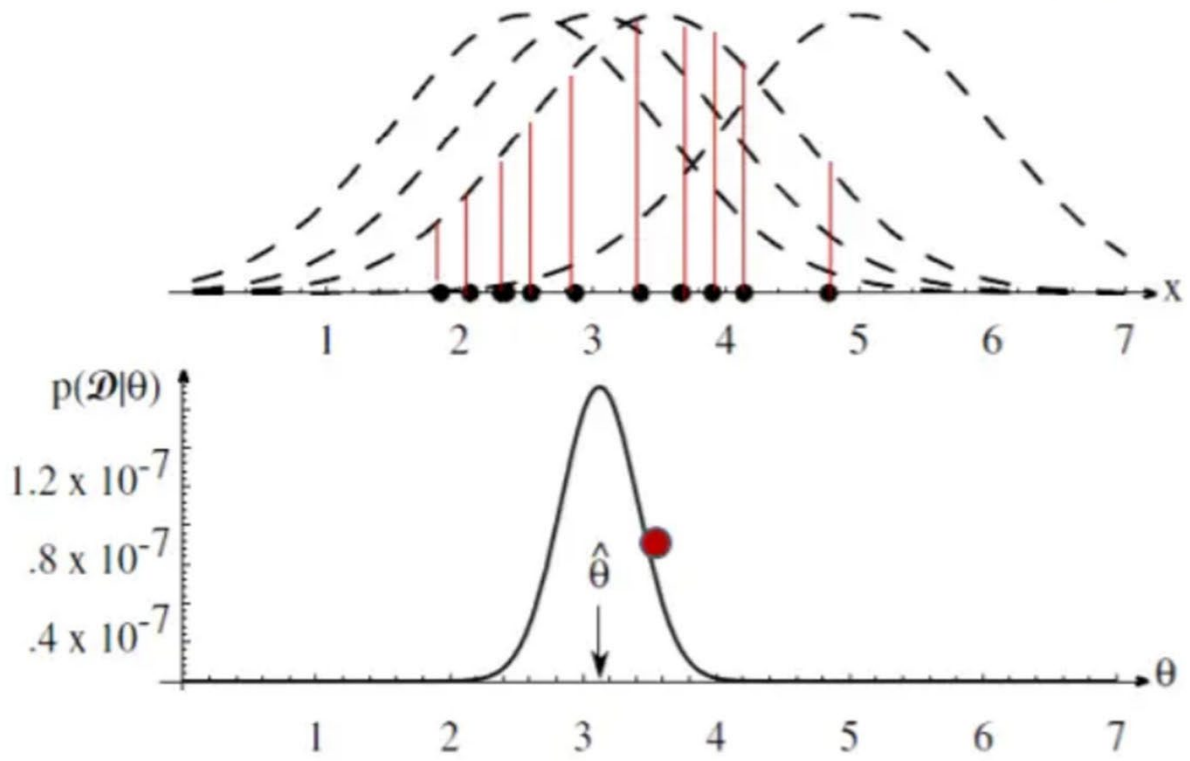
$$\widehat{\boldsymbol{\theta}}_{ML} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, p(\mathcal{D}|\boldsymbol{\theta})$$

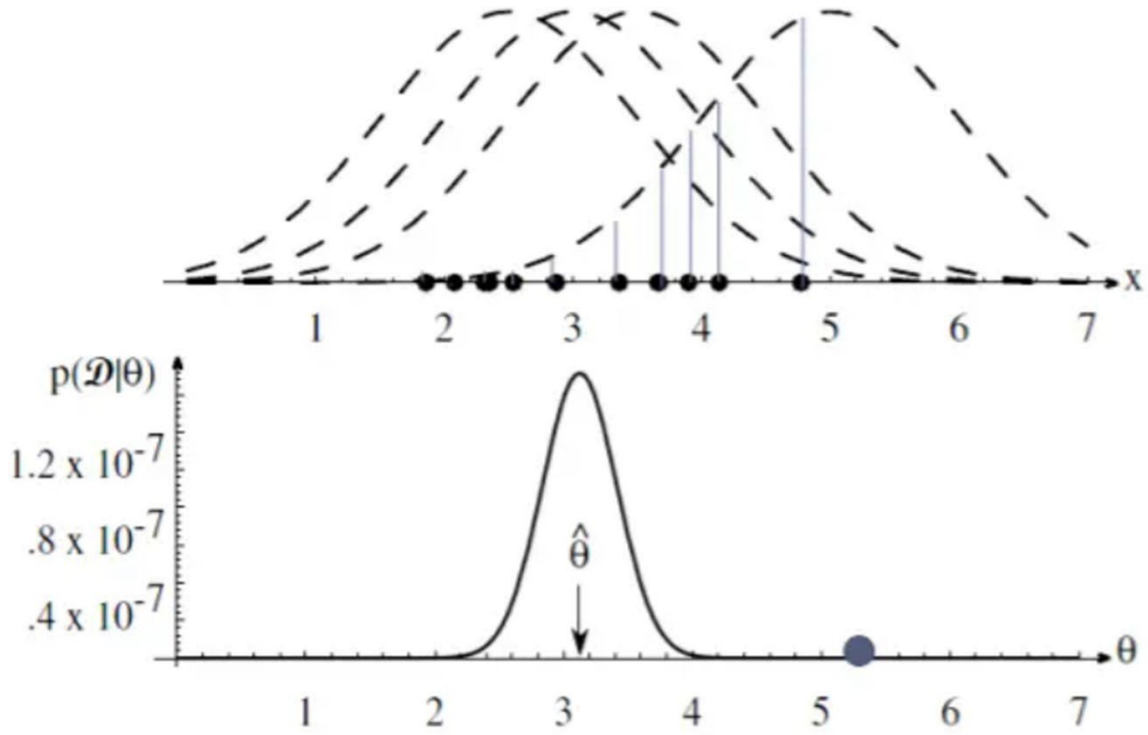
MLE



$\hat{\theta}$ best agrees with the observed samples

$$\widehat{\boldsymbol{\theta}}_{ML} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, p(\mathcal{D}|\boldsymbol{\theta})$$

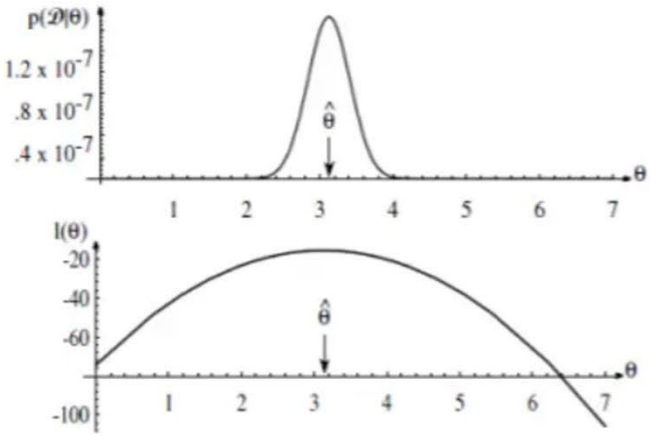

x
1
2
3
4
5
6
7
p(𝒟|θ)
1.2 x 10^-7
.8 x 10^-7
.4 x 10^-7
θ̂
θ
1
2
3
4
5
6
7


x
1
2
3
4
5
6
7
p(𝒟|θ)
1.2 x 10^-7
.8 x 10^-7
.4 x 10^-7
θ̂
θ
1
2
3
4
5
6
7

$$\mathcal{L}(\boldsymbol{\theta}) = \ln p(\mathcal{D}|\boldsymbol{\theta}) = \ln \prod_{i=1}^{N} p\left(\boldsymbol{x}^{(i)}|\boldsymbol{\theta}\right) = \sum_{i=1}^{N} \ln p\left(\boldsymbol{x}^{(i)}|\boldsymbol{\theta}\right)$$

$$\widehat{\boldsymbol{\theta}}_{ML} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, \mathcal{L}(\boldsymbol{\theta}) = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \sum_{i=1}^{N} \ln p\left(\boldsymbol{x}^{(i)}|\boldsymbol{\theta}\right)$$

Thus, we solve $\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \mathbf{0}$
to find global optimum



MLE:Bernoulli

Given: $\mathcal{D} = \{x^{(1)}, x^{(2)}, \ldots, x^{(N)}\}$, m heads (1), $N - m$ tails (0)

$$p(x|\theta) = \theta^x (1-\theta)^{1-x}$$

MLE:Bernoulli

Given: $\mathcal{D} = \{x^{(1)}, x^{(2)}, \ldots, x^{(N)}\}$, m heads (1), $N - m$ tails (0)

$$p(x|\theta) = \theta^x (1-\theta)^{1-x}$$

$$p(\mathcal{D}|\theta) = \prod_{i=1}^{N} p(x^{(i)}|\theta) = \prod_{i=1}^{N} \theta^{x^{(i)}} (1-\theta)^{1-x^{(i)}}$$

$$\ln p(\mathcal{D}|\theta) = \sum_{i=1}^{N} \ln p(x^{(i)}|\theta) = \sum_{i=1}^{N} \{x^{(i)} \ln \theta + (1-x^{(i)}) \ln(1-\theta)\}$$

$$\frac{\partial \ln p(\mathcal{D}|\theta)}{\partial \theta} = 0 \Rightarrow \theta_{ML} = \frac{\sum_{i=1}^{N} x^{(i)}}{N} = \frac{m}{N}$$

Logistic Regresion

$f(\boldsymbol{x}; \boldsymbol{w})$: probability that $y = 1$ given $\boldsymbol{x}$ (parameterized by $\boldsymbol{w}$)

$$P(y = 1|\boldsymbol{x}, \boldsymbol{w}) = f(\boldsymbol{x}; \boldsymbol{w})$$

$K = 2$
$y \in \{0,1\}$

$$P(y = 0|\boldsymbol{x}, \boldsymbol{w}) = 1 - f(\boldsymbol{x}; \boldsymbol{w})$$

$f(\boldsymbol{x}; \boldsymbol{w}) = \sigma(\boldsymbol{w}^T \boldsymbol{x})$
$0 \leq f(\boldsymbol{x}; \boldsymbol{w}) \leq 1$
estimated probability of $y = 1$ on input x

Example: Cancer (Malignant, Benign)

- $f(\boldsymbol{x}; \boldsymbol{w}) = 0.7$
- 70% chance of tumor being malignant

Logistic Regression

$f(\boldsymbol{x}; \boldsymbol{w})$ predicts posterior probabilities $P(y = 1|\boldsymbol{x})$

$$f(\boldsymbol{x}; \boldsymbol{w}) = \sigma(\boldsymbol{w}^T \boldsymbol{x})$$

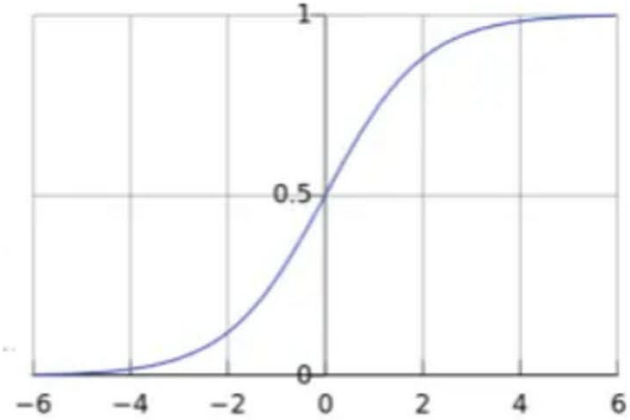
$\boldsymbol{x} = [1, x_1, \ldots, x_d]$
$\boldsymbol{w} = [w_0, w_1, \ldots, w_d]$

$\sigma(.)$ is an activation function

Sigmoid (logistic) function

- Activation function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



Decision surface

Decision surface $f(\boldsymbol{x}; \boldsymbol{w}) = \text{constant}$

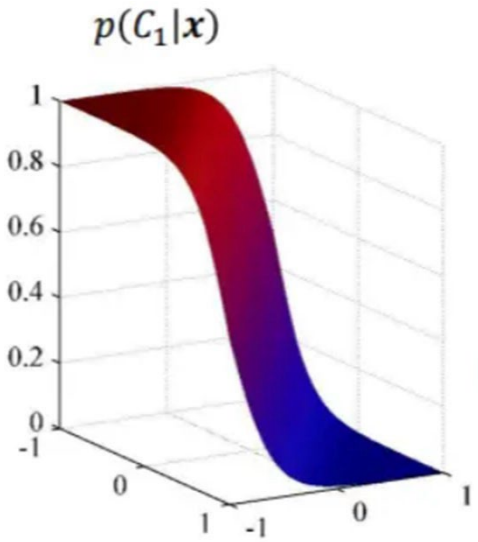
- $f(\boldsymbol{x}; \boldsymbol{w}) = \sigma(\boldsymbol{w}^T \boldsymbol{x}) = \frac{1}{1+e^{-(\boldsymbol{w}^T \boldsymbol{x})}} = 0.5$

Decision surfaces are linear functions of $\boldsymbol{x}$

if $f(\boldsymbol{x}; \boldsymbol{w}) \geq 0.5$ then $y = 1$
else $y = 0$

Equivalent to

if $\boldsymbol{w}^T \boldsymbol{x} + w_0 \geq 0$ then $y = 1$
else $y = 0$



[Bishop]

ML estimation

Maximum (conditional) log likelihood:

$$\hat{\boldsymbol{w}} = \underset{\boldsymbol{w}}{\operatorname{argmax}} \ \log \prod_{i=1}^{n} p\left(y^{(i)} | \boldsymbol{w}, \boldsymbol{x}^{(i)}\right)$$

$$p\left(y^{(i)} | \boldsymbol{w}, \boldsymbol{x}^{(i)}\right) = f\left(\boldsymbol{x}^{(i)}; \boldsymbol{w}\right)^{y^{(i)}} \left(1 - f\left(\boldsymbol{x}^{(i)}; \boldsymbol{w}\right)\right)^{(1-y^{(i)})}$$

$$\log p(\boldsymbol{y} | \boldsymbol{X}, \boldsymbol{w})$$
$$= \sum_{i=1}^{n} \left[y^{(i)} \log\left(f\left(\boldsymbol{x}^{(i)}; \boldsymbol{w}\right)\right) + (1 - y^{(i)}) \log\left(1 - f\left(\boldsymbol{x}^{(i)}; \boldsymbol{w}\right)\right) \right]$$

Logistic Regression: Loss function

$$\hat{\boldsymbol{w}} = \underset{\boldsymbol{w}}{\operatorname{argmin}} J(\boldsymbol{w})$$

$$J(\boldsymbol{w}) = -\sum_{i=1}^{n} \log p\left(y^{(i)} \middle| \boldsymbol{w}, \boldsymbol{x}^{(i)}\right)$$

$$= \sum_{i=1}^{n} -y^{(i)} \log\left(f\left(\boldsymbol{x}^{(i)}; \boldsymbol{w}\right)\right) - (1 - y^{(i)}) \log\left(1 - f\left(\boldsymbol{x}^{(i)}; \boldsymbol{w}\right)\right)$$

No closed form solution for

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = 0$$

However $J(\boldsymbol{w})$ is convex.

Logistic Regression: Gradient descent

$$\boldsymbol{w}^{t+1} = \boldsymbol{w}^{t} - \eta \nabla_{\boldsymbol{w}} J(\boldsymbol{w}^{t})$$

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \sum_{i=1}^{n} \left(f\left(\boldsymbol{x}^{(i)}; \boldsymbol{w}\right) - y^{(i)} \right) \boldsymbol{x}^{(i)}$$

Is it similar to gradient of SSE for linear regression?

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \sum_{i=1}^{n} \left(\boldsymbol{w}^{T} \boldsymbol{x}^{(i)} - y^{(i)} \right) \boldsymbol{x}^{(i)}$$

Multi-class logistic regresion

$K > 2$
$y \in \{1,2,\dots,K\}$

$$f_k(\boldsymbol{x};\boldsymbol{W}) = p(y = k|\boldsymbol{x}) = \frac{\exp(\boldsymbol{w}_k^T\boldsymbol{x})}{\sum_{j=1}^{K}\exp(\boldsymbol{w}_j^T\boldsymbol{x})}$$

- Normalized exponential (aka softmax)
  - If $\boldsymbol{w}_k^T\boldsymbol{x} \gg \boldsymbol{w}_j^T\boldsymbol{x}$ for all $j \neq k$ then $p(C_k|\boldsymbol{x}) \simeq 1, p(C_j|\boldsymbol{x}) \simeq 0$

$$p(C_k|\boldsymbol{x}) = \frac{p(\boldsymbol{x}|C_k)p(C_k)}{\sum_{j=1}^{K}p(\boldsymbol{x}|C_j)p(C_j)}$$

For each class k, $f_k(\boldsymbol{x}; \boldsymbol{W})$ predicts the probability of $y = k$

- i.e., $P(y = k|\boldsymbol{x}, \boldsymbol{W})$

On a new input $\boldsymbol{x}$, to make a prediction, pick the class that maximizes $f_k(\boldsymbol{x}; \boldsymbol{W})$:

$$\alpha(\boldsymbol{x}) = \underset{k=1,\ldots,K}{\operatorname{argmax}} f_k(\boldsymbol{x})$$

if $f_k(\boldsymbol{x}) > f_j(\boldsymbol{x}) \quad \forall j \neq k$ then
decide C_k